

Phase 3: Model Development

3.1 Introduction

The Model Development phase is the core of the Smart Waste Management System. In this phase, all system components are designed and implemented, including data processing, route optimization, prediction models, and the user interface.

The system is developed using a modular approach, ensuring scalability, maintainability, and ease of integration with real-world IoT devices in the future.

3.2 System Architecture Overview

The system follows a layered architecture consisting of:

1. Data Layer (Database)
2. Application Layer (Backend – Flask)
3. Logic Layer (Algorithms & Models)
4. Presentation Layer (Frontend Dashboard)

Each layer interacts with others to ensure smooth data flow and real-time system functionality.

3.3 Modules of the System

3.3.1 Sensor Simulation Module

File:

- sensors/simulator.py

Description:

This module simulates IoT sensors by generating random data for garbage bins.

Functions:

- Generate fill level (0–100%)
- Generate garbage weight
- Assign bin status

Purpose:

- Replace real sensors
 - Provide continuous data input
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3.3.2 Backend Module

File:

- app.py

Description:

This is the main controller of the system built using Flask.

Responsibilities:

- Handle API requests
- Process incoming data
- Communicate with database
- Send responses to frontend

Key Features:

- REST API endpoints
 - Real-time data handling
 - Integration with simulation module
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3.3.3 Database Module

File:

- database/db_setup.py

Description:

Manages data storage using SQLite.

Responsibilities:

- Create tables
 - Store sensor data
 - Retrieve historical data
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3.3.4 Frontend Module

Files:

- templates/index.html
- static/css/style.css
- static/js/dashboard.js

Description:

Provides a user interface for monitoring bins.

Features:

- Display garbage levels
- Show alerts

- Update data dynamically
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3.4 Route Optimization Model

3.4.1 Algorithm Used

- Nearest Neighbor Algorithm

3.4.2 Purpose

To determine the most efficient path for waste collection vehicles.

3.4.3 Working Principle

1. Start from a location
 2. Visit nearest bin
 3. Continue to next closest bin
 4. Repeat until all bins are covered
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3.4.4 Advantages

- Simple to implement
 - Fast execution
 - Suitable for small datasets
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3.5 Prediction Model

3.5.1 Algorithm Used

- Linear Regression
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3.5.2 Purpose

To predict future garbage levels based on historical data.

3.5.3 Working Process

1. Collect historical fill level data
 2. Train regression model
 3. Predict future values
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3.5.4 Benefits

- Helps in planning collection schedules
 - Prevents overflow
 - Improves efficiency
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3.6 Data Flow in the System

1. Sensor simulator generates data
 2. Backend receives data
 3. Data stored in database
 4. Frontend requests data
 5. System:
 - Displays data
 - Runs prediction model
 - Optimizes routes
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3.7 API Design

Example Endpoints:

- /get_data → Fetch bin data
 - /add_data → Insert sensor data
 - /optimize_route → Get best route
 - /predict → Get predicted fill levels
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3.8 User Interface Design

Dashboard Features:

- Real-time bin monitoring
- Alerts for full bins
- Data visualization

UI Goals:

- Simple design
 - Easy navigation
 - Clear data representation
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3.9 Integration of Modules

All modules are integrated as follows:

- Simulator → Backend → Database
- Database → Backend → Frontend
- Backend → Optimization & Prediction Models

This ensures smooth system functionality.

3.10 Testing During Development

- Tested random data generation
 - Verified database storage
 - Checked API responses
 - Validated UI updates
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3.11 Challenges Faced

- Handling continuous data updates
 - Managing database efficiently
 - Simulating realistic data
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3.12 Solutions Implemented

- Used structured modules
 - Optimized database queries
 - Implemented clean API design
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3.13 Advantages of the Model

- Modular and scalable
 - Easy to upgrade
 - Supports real-time processing
 - Ready for IoT integration
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3.14 Limitations

- Uses simulated data
- Basic algorithms used
- Limited real-world complexity

3.15 Future Enhancements

- Use advanced ML models
 - Integrate real-time sensors
 - Improve route optimization using AI
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3.16 Conclusion

The Model Development phase successfully integrates all components of the system, including simulation, backend, database, and frontend. It provides a working prototype of a smart waste management system capable of real-time monitoring, prediction, and optimization.

This phase forms the backbone of the entire project and ensures that the system is functional, efficient, and scalable.